

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]

Do not open this Test Booklet until you are asked to do so.
Read carefully the Instructions on this Test Booklet.

PAPER-2

Paper : Physics, Chemistry & Mathematics

- Important Instructions :**
1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
 2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75** questions. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **25 questions** in each subject and each subject having **Two sections**.

(i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.

(ii) Section-II contains 5 **Numerical Value Type** questions
Marking scheme : +4 for correct answer and 0 in all other cases.
 6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
 7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
 8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
 9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
 10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

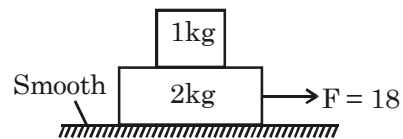
ALLEN
PART 1 - PHYSICS

SECTION-I : (Maximum Marks : 80)

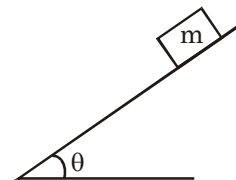
- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. In a new system of units called royal system of units, units of length, mass and time are each ten times of respective SI units. The unit of force in this system will be
 (A) equal to that in SI system
 (B) 10 times that of SI system
 (C) 100 times that of SI system
 (D) 1000 times that of SI system
2. A particle moves 3m in north-east direction then 4m in upward direction and then 12m in north-west direction. Find the net displacement
 (A) 13 m (B) 11 m
 (C) 14 m (D) none of these

3. A constant force acting on a particle of mass 2kg is given by $(\vec{F} = 2\hat{i} + \hat{j} - \hat{k})\text{N}$. What is the work done by given force during motion of particle from point A (1m, 1m, 0) to point B (3m, 4m, 2m) ?
 (A) 5 J (B) 2 J (C) 4 J (D) 6 J
4. In the figure, there is no relative motion between the two blocks. Force of friction acting on 1kg block is :-

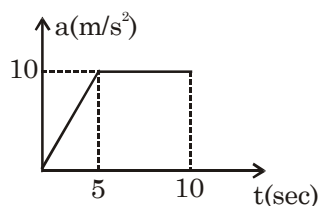


- (A) Zero (B) 3N (C) 6N (D) 4N
5. In the given figure the coefficient of friction between block and incline is μ . If the block slides on inclined with constant velocity, the friction acting on block will be :-



- (A) $\mu mg \sin \theta$ (B) $\mu mg \tan \theta$
 (C) $mg \sin \theta$ (D) $> \mu mg \cos \theta$

6. If at $t = 0$, velocity of particle is 2 m/s , find velocity at $t = 10 \text{ sec}$.



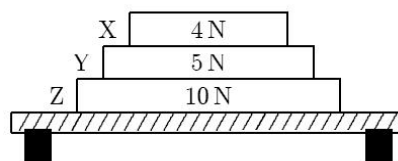
- (A) 100 m/s (B) 73 m/s
(C) 77 m/s (D) 70 m/s
7. A uniform heavy rope is being pulled by a force 100 N , find tension at its' mid point :-



- (A) 25 N (B) 50 N
(C) 75 N (D) 100 N
8. A body of mass 1 kg starts from origin with initial velocity $\vec{u} = (8\hat{i} + 10\hat{j}) \text{ m/s}$. If a constant force $\vec{F} = (-2\hat{i} - 3\hat{j}) \text{ N}$ acts on the body, find the time in which x-component of velocity becomes zero :-

- (A) 8 sec (B) 2 sec
(C) 4 sec (D) $\frac{10}{3} \text{ sec}$

9. Three books (X, Y, and Z) rest on a table. The weight of each book is indicated. The net force acting on book Y is:

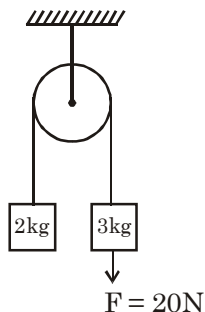


- (A) 4 N down (B) 5 N up
(C) 9 N down (D) zero
10. Two cars moving towards each other having 30 km/hr and 40 km/hr constant speed. Initially distance between cars is 70 km then time after which they meet, is

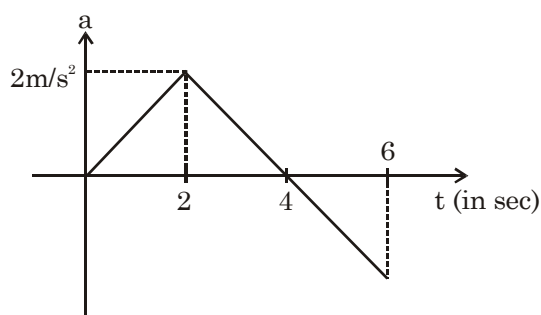


- (A) 1 sec (B) 1 hour
(C) 60 sec (D) 120 sec
11. A ball is projected in a vertical plane with an initial velocity 10 m/s at an angle 30° above the horizontal level. When the velocity v makes 45° in downwards from the horizontal level, the value of v^2 is :
- (A) $10 \text{ m}^2/\text{s}^2$ (B) $50 \text{ m}^2/\text{s}^2$
(C) $100 \text{ m}^2/\text{s}^2$ (D) $150 \text{ m}^2/\text{s}^2$

12. A lift is moving with constant velocity. A coin is dropped at height 20m from the floor of lift, find the time taken by coin to reach the floor :- (Take $g = 10 \text{ m/s}^2$)
 (A) 1 s (B) 0.5 s (C) 2 s (D) 1.5 s
13. Acceleration of block of mass 2 kg will be : (Take $g = 10 \text{ m/s}^2$)



- (A) 2 m/s^2 (B) 6 m/s^2 (C) 4 m/s^2 (D) 5 m/s^2
14. a-t graph of a particle is shown in the figure below. If at $t = 0$, $v = 0$ the maximum speed of the particle will be at time?



- (A) $t = 1 \text{ sec}$ (B) $t = 4 \text{ sec}$
 (C) $t = 6 \text{ sec}$ (D) $t = 3 \text{ sec}$

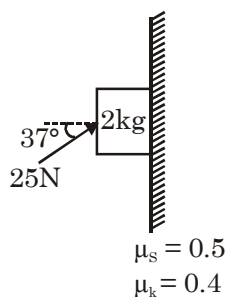
15. The length of a seconds hand in a watch is 1 cm. The magnitude of change in velocity of its tip in 15 second is :
 (A) zero
 (B) $\pi/30 \sqrt{2} \text{ cm/s}$
 (C) $\pi/30 \text{ cm/s}$
 (D) $\pi \sqrt{2} / 30 \text{ cm/s}$
16. What is the average angular speed of the second hand on a clock (in rad/s) ?
 (A) 6.28
 (B) 0.105
 (C) 0.0167
 (D) 1.745×10^{-3}
17. A mass of 4kg is moving along a circular path of radius 2m. If mass moves with angular speed of 4 radian/sec, it's kinetic energy will be :
 (A) 60 J
 (B) 128 J
 (C) 80 J
 (D) 54 J
18. If a particle moves in a circle of radius 4m at speed given by $v = 2t$ where v in m/s & t in sec then at $t = 4 \text{ sec}$, centripetal acceleration of particle will be :
 (A) 16 m/s^2
 (B) 32 m/s^2
 (C) 64 m/s^2
 (D) 4 m/s^2

19. If a particle moving with constant angular acceleration in a circular path changes its angular velocity from 8 radian/sec to 12 radian/sec in 4 sec, find angular displacement of particle

(A) 20 radian
(B) 40 radian
(C) 80 radian
(D) 10 radian

20. A 25 N force is applied to the block as shown in figure. Choose the **CORRECT** option if initial velocity of block is zero

(Take $g = 10 \text{ m/s}^2$) :-



- (A) Normal contact force is 10N.
(B) Friction force is 8N.
(C) Friction force is 10N.
(D) Block is in equilibrium.

SECTION-II : (Maximum Marks: 20)

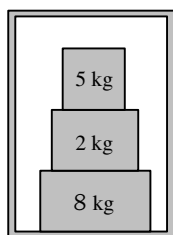
- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

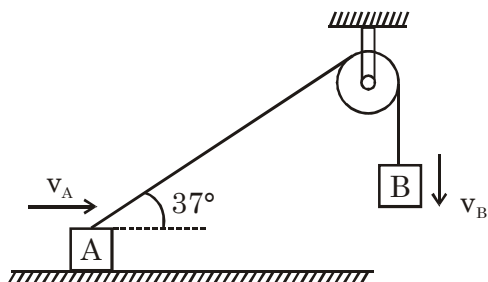
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●	●	0	0	0	0	0	0	0	0	●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Three boxes are placed in a lift. When acceleration of the lift is 1 m/s^2 , the net force (in N) on the 5 kg box is:

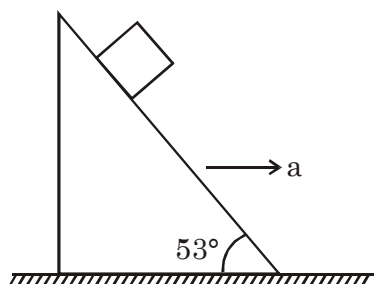


2. If velocity of block A is 5 m/s . Find velocity of block B in m/s .



3. If acceleration (in m/s^2) of wedge (rightwards) so that block does not move w.r.t. wedge is $\frac{8n}{3}$. Find value of n .

(Take $g = 10 \text{ m/s}^2$)



4. A cyclist speeding at 2 m/s on a level road takes a sharp circular turn of radius 2 m without reducing the speed. If minimum coefficient of static friction between tyres and road so that cyclist does not slip while

taking the turn is $\frac{1}{n}$, find value of n .

(Take $g = 10 \text{ m/s}^2$)

5. If time to cross a 40 m wide river is 5 sec with zero drift. If velocity of boat in still water is 10 m/s , find velocity (in m/s) of river flow.

ALLEN
PART 2 - CHEMISTRY

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

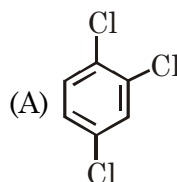
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

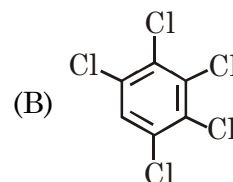
Negative Marks : -1 In all other cases

1. Total number of sulphur atoms in $H_2S_nO_6$ which form covalent bond in their ground state :-
 (A) n (B) n-1
 (C) n-2 (D) n+1
2. **CORRECT** order of number of planes present in molecule which contain maximum number of atoms of corresponding molecule :-
 (A) $IF_7 > SF_6 > PCl_5$ (B) $PCl_5 > SF_6 > IF_7$
 (C) $SF_6 > IF_7 > PCl_5$ (D) $PCl_5 > IF_7 > SF_6$

3. Comment on the C – C bond length for C_2H_6 and C_2F_6 compounds :
 (A) $d_{C-C}(C_2H_6) > d_{C-C}(C_2F_6)$
 (B) $d_{C-C}(C_2F_6) > d_{C-C}(C_2H_6)$
 (C) $d_{C-C}(C_2F_6) = d_{C-C}(C_2H_6)$
 (D) Can't be predicted
4. Choose the **CORRECT** statement.
 (A) ClF_2^- is linear while ClF_2^+ is bent
 (B) All bond angles in CH_2Cl_2 are not identical
 (C) O–O bond length : $H_2O_2 > O_2F_2$
 (D) All are correct.
5. Which of the following molecule exist?
 (A) BF_6^{3-} (B) $SiCl_6^{2-}$
 (C) XeH_2 (D) SH_2
6. Among the following which does not have same value of dipole moment as C_6H_5Cl .



(C) m- $C_6H_4Cl_2$



(D) p- $C_6H_4Cl_2$

7. In which of the following molecules, the central atom uses pure atomic orbitals for bond formation?
 (A) SO_3 (B) AsF_3
 (C) CO_2 (D) A & C both
8. Which of the following is **INCORRECT** ?
 (A) $\text{C} < \text{N} < \text{O}$ (2nd IE)
 (B) $\text{Li}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$ (size)
 (C) Diamond is hardest natural substance
 (D) For $\text{A}_{(\text{g})} + \text{e}^- \longrightarrow \text{A}_{(\text{g})}^-$ ΔH must be negative
9. Which of the following has the maximum number of unpaired electrons ?
 (A) Mg^{2+} (B) Ti^{3+}
 (C) V^{3+} (D) Fe^{2+}
10. Which of the following species does not involve d-orbital in hybridisation of central atom ?
 (A) ICl_3 (B) $\text{PCl}_5(\text{g})$
 (C) I_3^- (D) NO_3^-
11. At a certain temperature and pressure, 2 moles of CO_2 gas occupies 50 litre, then at same temperature and pressure, volume occupied by 0.5 mole of He will be:
 (A) 25 L (B) 250 L
 (C) 125 L (D) 12.5 L
12. In a gaseous mixture, if 0.5 gm of He, 20 gm of SO_3 and 3.5 gm of N_2 is present, then the average molar mass of mixture is :
 (A) 20 gm (B) 24 gm
 (C) 12 gm (D) 48 gm
13. If vapour density of a gas is 13, then the volume occupied by 20.8 gm of this gas at STP will be:
 (A) 22.7 L
 (B) 18.16 L
 (C) 17.92 L
 (D) 36.32 L
14. Which of the following contains maximum number of carbon atoms:
 (A) 1.6 gm of CH_4
 (B) 8.4 gm of C_3H_6
 (C) 24 gm of CH_3COOH
 (D) 18.4 gm HCOOH
15. Atoms present in 3.2 gm of Sulphur is equal to the number of atoms present in -
 [Given : S = 32, Mg = 24, Na = 23, C = 12]
 (A) 2.4 gm of magnesium
 (B) 1.2 gm of carbon
 (C) 2.3 gm of sodium
 (D) all of the above

16. If 40.6 gm of a hydrocarbon contains 4.21×10^{23} molecules, then the hydrocarbon can be
 (A) C_2H_2 (B) C_3H_8
 (C) C_4H_{10} (D) C_2H_6
17. Number of SO_2 molecules in 3.2 gm of it is equal to:
 (A) Molecules in 8 gm of oxygen gas
 (B) Molecules in 0.9 gm of H_2O
 (C) Atoms in 1 gm of He
 (D) Atoms in 1 gm of H_2 gas
18. The dissociation of N_2O_4 takes place as:
 $N_2O_4 \longrightarrow 2NO_2$
 If degree of dissociation is 0.2, then the average molar mass will be :
 (A) 92 (B) 69.66
 (C) 76.66 (D) 55.2
19. The weight of 1 atom of element 'X' is 'Y' amu, then gram molecular mass of X_2 is:
 (in gram)
 (A) $2Y N_A$ (B) Y / N_A
 (C) Y (D) 2Y
20. If 0.5 mole $BaCl_2$ is mixed with 0.2 mole of Na_3PO_4 . The maximum moles of $Ba_3(PO_4)_2$ formed is .
 (A) 0.7 (B) 0.5/3
 (C) 0.3 (D) 0.1

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

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●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

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●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Mass (in gm) of SO_3 which contains 2 moles of oxygen atoms is.
2. If from a container containing 3.4 gm of NH_3 , 4×10^{22} molecules of NH_3 are removed, then what % of molecules are left in the container? ($N_A = 6 \times 10^{23}$)
3. How many of following has/have X-X bond (bond present between two same atoms)
 $\text{H}_2\text{S}_2\text{O}_7$, $\text{H}_2\text{S}_2\text{O}_3$, $\text{H}_2\text{S}_2\text{O}_8$, $\text{H}_4\text{P}_2\text{O}_6$
 (hypophosphoric acid), $\text{H}_2\text{S}_2\text{O}_5$, $\text{H}_2\text{N}_2\text{O}_2$, SiO_2 , CrO_5
4. Consider the following molecules :-
 XeO_3 , XeF_5^- , SO_2 , I_3^+ , IF_3 , BrF_5 , IF_6^+ , XeO_3F_2 , $\text{N}(\text{CH}_3)_3$
 x = number of polar species
 y = species having atleast one $\text{p}_\pi\text{-d}_\pi$ bond
 Find the value of $(x \times y)$
5. How many of following pair/s present below in which first species has lower boiling point than second species.
 (i) $\text{C}_2\text{H}_5\text{Cl}$, $\text{C}_2\text{H}_5\text{OH}$
 (ii) HCl , HF
 (iii) NH_3 , H_2O
 (iv) He , H_2
 (v) 1,3,5-trichlorobenzene, 1,2,3-trichlorobenzene
 (vi) o-nitrophenol, p-nitrophenol
 (vii) BF_3 , BMe_3

ALLEN
PART 3 - MATHEMATICS

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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1. If $\sin^2\theta - 3\sin\theta + 2 = 0$ where $\theta \in (0, 2\pi)$ number of possible values of θ is
 (A) 1 (B) 2 (C) 0 (D) 4
2. If $ax^2 + bx + c = 0$ has roots α and β then $\frac{\alpha + \beta}{\alpha\beta}$ is
 (A) $-\frac{b}{c}$ (B) $-b$ (C) $-\frac{c}{b}$ (D) c
3. The number of different values of θ satisfying the equation $\cos\theta + \cos 2\theta = -1$ and at the same time satisfying the condition $0 < \theta < 360^\circ$ is
 (A) 1 (B) 2 (C) 3 (D) 4

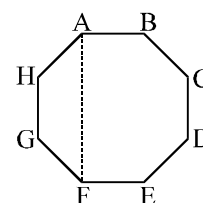
4. If $x = 1$ and $x = 2$ are solutions of the equation $x^3 + ax^2 + bx + c = 0$ and $a + b = 1$, then value of b is
 (A) -4 (B) 5 (C) 2 (D) 1

5. $\tan \theta = \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots \infty}}}$ where $\theta \in (0, 2\pi)$

number of value(s) of θ is

- (A) 1 (B) 2 (C) 3 (D) 0

6. The figure (not drawn to scale) shows a regular octagon ABCDEFGH with diagonal $AF = 1$. The numerical value of the side of the octagon is



- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{2} - 1$ (D) 2

7. Let $S = \sec 0 + \sec \frac{\pi}{5} + \sec \frac{2\pi}{5} + \sec \frac{3\pi}{5} + \sec \frac{4\pi}{5}$ and $P = \tan \frac{\pi}{9} \cdot \tan \frac{2\pi}{9} \cdot \tan \frac{4\pi}{9}$ if $S + P = a\sqrt{a} \sin \frac{7\pi}{12}$, then 'a' is
 (A) 2 (B) 3 (C) 4 (D) 5

8. Number of solution(s) of the equation

$$x^{0.5 \log_{\sqrt{x}}(x^2-x)} = 3^{\log_9 4} \text{ is}$$

- (A) 1 (B) 0 (C) 2 (D) 4

9. Range of the expression

$$y = \frac{x^2 - 2x - 8}{x^2 - 4x - 5}, \text{ for all permissible value of}$$

x, is

- (A) $[-2, 2]$ (B) $[2, \infty]$
(C) $(-\infty, 2]$ (D) $(-\infty, \infty)$

10. Quadratic equation with rational coefficients if one of its root is $\cot^2 18^\circ$, is

- (A) $x^2 - 5x + 10 = 0$ (B) $x^2 - 10x + 5 = 0$
(C) $x^2 + 10x - 5 = 0$ (D) $x^2 - 10x - 5 = 0$

11. Let α and β be the roots of the quadratic equation $(x-2)(x-3)+(x-3)(x+1)+(x+1)(x-2)=0$. The value of

$$\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)} + \frac{1}{(\alpha-3)(\beta-3)}, \text{ is}$$

- (A) 0 (B) 1 (C) 2 (D) 10

12. The sum of the first five terms of a geometric series is 189, the sum of the first six terms is 381, and the sum of the first seven terms is 765. The common ratio in this series is

- (A) 2 (B) $\frac{1}{2}$ (C) 3 (D) $\frac{1}{3}$

13. If b is the arithmetic mean between a and x ; b is the geometric mean between ' a ' and y ; ' b ' is the harmonic mean between a and z , ($a, b, x, y, z > 0$) then the value of xyz is

- (A) a^3 (B) b^3
(C) $\frac{b^3(2a-b)}{2b-a}$ (D) $\frac{b^3(2b-a)}{2a-b}$

14. If a, b and c are numbers for which the equation

$$\frac{x^2 + 10x - 36}{x(x-3)^2} = \frac{a}{x} + \frac{b}{x-3} + \frac{c}{(x-3)^2} \text{ is an}$$

identity, then $a + b + c$ equals

- (A) 2 (B) 3 (C) 10 (D) 8

15. In a $\triangle ABC$, $\angle B = \frac{\pi}{3}$ and $\angle C = \frac{\pi}{4}$. Let D divide BC internally in the ratio $1 : 3$, then

$\frac{\sin(\angle BAD)}{\sin(\angle CAD)}$ is equal to

- (A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{\sqrt{2}}{3}$

16. At a point 15 metre away from the base of a 15 metre high house, the angle of elevation of the top is

- (A) 45° (B) 30° (C) 60° (D) 90°

17. If $x, y, z > 0$ and $x + y + z = 1$, then

$\frac{xyz}{(1-x)(1-y)(1-z)}$ is necessarily

- (A) ≥ 8 (B) $\leq \frac{1}{8}$
(C) $< \frac{1}{8}$ (D) None of these

18. If a, b, c are in A.P. then $\sin A, \sin B, \sin C$ must be in (where all symbols in their usual notation for ΔABC)

- (A) A.P. (B) G.P.
(C) H.P. (D) None of these

19. If $x^2 - 1 < 0$ then number of integral values of x equal to

- (A) 3 (B) 0
(C) 1 (D) 2

20. If $\frac{29}{12}$ can be expressed as $a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}}$,

($a, b, c, d \in \mathbb{N}$) then find the value of $a^3 + b^3 + c^3 + d^3$ is equal to

- (A) 4 (B) 16
(C) 11 (D) 32

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Positive integers p, q, r, s satisfying $\tan \frac{\pi}{24} = (\sqrt{p} - \sqrt{q})(\sqrt{r} - s)$ then $\frac{p+q+r+s}{4}$ is
- A group of members know at least one of the two languages, Hindi or Kannada. In the group, 150 members know Hindi and 80 members know Kannada and 70 of them know both. Number of members in the group is ?
- The expression $\sin^2 \theta + \sin^2(120^\circ + \theta) + \sin^2(120^\circ - \theta)$ remains constant $\forall \theta \in \mathbb{R}$. The value of this constant is
- One root of $mx^2 - 10x + 3 = 0$ is two third of the other root. The sum of the roots equals to
- Maximum value of $f(x, y) = 7x^2 + 4xy + 3y^2$ subjected to $x^2 + y^2 = 1$, is

